CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CL) Examination

November 2012

CL310 Irrigation & Hydraulic Structures

Date: 09.11.2012, Friday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5.

SECTION – I

Q - 1 (a) Design an unlined irrigation channel for alluvial soil by Kennedy's theory for following data: **[07]**
Discharge = 45 cumecs, Channel bed slope = 1:5000, N = 0.0225, m = 1.05, Side slope of trapezoidal channel = 0.5 :1.

Q - 2 (a) Draw a neat sketch of Chimney drain in embankment dam. **[02]**

(b) What are the various modes of failure of a gravity dam? Discuss each of them briefly. **[07]**

(c) What is meant by an energy dissipator? Discuss the various methods used for energy dissipation below spillways. **[07]**

OR

Q - 2 (a) Draw a neat sketch of cross-section of rockfill dam. **[02]**

(b) Discuss the various causes of failure of embankment dam. **[07]**

(c) Discuss briefly the various types of energy dissipators that are used for energy dissipation below overflow spillways, under different relative positions of T.W.C and J.H.C. **[07]**

Q - 3 (a) Distinguish between high and low gravity dam with neat sketches. **[07]**
Find the limiting height of the low gravity dam with the following data: allowable compressive strength of concrete = 500 N/cm^2 and specific gravity of concrete is 2.41.

(b) Draw neat sketches of following showing all the components: **[05]**
(i) Ogee spillway with vertical u/s face
(iii) Chute spillway

OR

Q - 3 (a) Draw neat sketches showing all the components of following type of Earthen dam. **[07]**
(i) Homogeneous Type (ii) Zoned Type (iii) Diaphragm Type

(b) What is a spillway gate and what are the merits and demerits of installing such gates? **[05]**

SECTION – II

Q - 4 (a) Define following terms: **[03]**

Kor period, Kor depth, Gross commanded area (G.C.A)

(b) Give the detailed classification of methods of irrigation. **[02]**

Q - 5 (a) The base period, intensity of irrigation and duty of water for various crops under a canal system are given in the table below. Determine the reservoir capacity if the culturable commanded area is 50,000 hectares, canal losses are 20% and reservoir losses are 10%. **[7]**

Crop	Base period (days)	Duty of water at the field (hectares/cumec)	Intensity of irrigation (%)
Vegetables	130	700	10
Sugarcane	350	1700	20
Cotton	200	1400	10
Rice	150	800	15
Wheat	120	1800	20

(b) Define canal fall. Where are canal falls exactly located? **[06]**

OR

Q - 5 (a) Determine the design discharge of the canal for the following data: **[07]**

Crop	Base Period (days)	Area of crop (ha)	Duty at the head (hectares/cumec)
Sugarcane	365	250	650
Paddy (Kharif)	115	1800	750
Bajra (kharif)	110	2000	1200
Wheat (Rabi)	110	2500	1000
Vegetable(HW)	985	1500	650

Take time factor = 0.7, capacity factor = 0.8 and conveyance efficiency= 65 %

(b) What are the criteria taken into consideration for selection of cross drainage works? **[06]**

Q - 6 (a) Compare bligh's theory and khosla's theory for hydraulic structure. **[06]**

(b) Draw a neat layout of diversion head-works and indicate the various components of the system. **[06]**

(c) Write a short note on Sub-surface Irrigation Methods. **[05]**

OR

Q - 6 (a) Define: creep length. Explain Bligh's creep theory for the design of impervious floor on permeable soils. Also, discuss the limitations of this theory. **[06]**

(b) Distinguish between weir and barrage. **[06]**

(c) How the quality of water affect on irrigation methods. **[05]**

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